CHENGHAO XU

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EDUCATION

Southern University of Science and Technology Shenzhen, China

- B.Eng in Mechanical Engineering *Sep.2016 - Jun.2020*
  Overall GPA: **3.65** / 4.00, Major GPA: **3.85** / 4.00
- *Courses: CAD and Engineering Drawing, Fundamentals of Machine Design, Advanced Manufacturing Practice*

Delft University of Technology Delft, Netherlands

- MSc Robotics (GPA: 8.2/10) *Sep.2020 - Present*
- *Courses: Dynamics & Control, Planning & Decision Making, Machine Perception, Deep Learning*

The University of British Columbia Vancouver, Canada

- Vancouver Summer Program (Grades: A-) *Jul.2017 - Aug.2017*

PROFESSIONAL EXPERIENCE

Multi-Camera Real-Time Surveillance Video Stitching
Computer Vision Engineer *.Internship*
Lely Technologies, Netherlands *Feb.2022 – Present*

- Investigated feature-based image stitching methods for multiview images with parallax, and conducted a comparative analysis of **ORB, BRISKS, and SIFT features** for video stitching
- Proposed a feature matching strategy based on the AutoStitch framework by matching the features within corresponding ROIs separately to improve overall stitching performance and robustness

Control Engineer for Industrial Manipulator *Internship*
ROKAE Robotics, China *Jul.2021 - Sep.2021*

- Developed the online trajectory planning algorithm for industrial robots based on the **discrete-time double S profile**, which achieves trajectory update in real-time with **continuous acceleration**
- Accomplished trajectory planning and optimization under multiple industrial scenarios, and simulated joint states (such as peak torque, angular speed, power) during the movements using Robotics System Toolbox
- Investigated the impact of DH parameter deviation on position accuracy of the end effector by Robotics Toolbox

Quadruped Robot based on Discontinuous Terrain Perception
Research Assistant *Advisor: Dr. Jianwen Luo*
SUSTech Institute of Robotics, China *Jun.2019 - Dec.2019*

- Accomplished and optimized the design of robot structure by SolidWorks to obtain higher stability
- Constructed the prototype and adjutsted motor control to achieve basic stepping or walking locomotion
- Secured funding from Youth Program of National Natural Science Foundation of China

Upper-Limb Passive Adaptive Assisted Exoskeleton
Research & Development Intern *Advisor: Prof. Hongqiang Wang*
Milebot Robotics, China . *Jun.2019 - Sep.2019*

- Conducted marketing analysis on exoskeleton products and used decision matrix to generate design concepts
- Optimized structural design of the energy storage and adaptive device for achieving adjustable assist effect
- Accomplished assist effect through mechanical analysis and numerical simulations using MATLAB
- Investigation of EMG signal and oxygen consumption for their secondary effect in the working environment

PROJECT EXPERIENCE

Path Planner for Quadrotor based on RRT* and k-PRM Methods
Course Project *Nov.2021 - Jan.2022*

- Employed RRT* and k-PRM path planner to generate collision-free path on 3D random obstacle map

- Accomplished trajectory optimization using minisnap and corridor bounding method

Manufacture of Aluminum Hollow Nested Cubes

Course Project *Feb.2019 - May.2019*

- Accomplished the structural design and machining simulation of aluminum nested cubes
- Delivered the process planning with the skilled operation of CNC machining center and wire-cut EDM

PUBLICATIONS & PRESENTATIONS

- "Design and Verification of Passive Upper Arm Assisted Exoskeleton"**(Poster)** Presented at Annual Conference of Chinese Robotics Society 2019.
- "Breaking the Wall of Intensive Work Above Head" Presented at Falling Walls Lab 2019 in Berlin, Germany.
- Patent: Chen, L., Liu, H., **Xu, C.**, etc. Assisted exoskeleton devices. (Appl. No. 201921545381.6)
- Patent: Chen, L., Liu, H., **Xu, C.**, etc. Assisted exoskeleton devices. (Appl. No. 201910870782.7)

AWARDS & HONORS

- 2016 Excellence Award | *Freshmen Scholarship*
- 2016 Champion | *The 2nd Shenzhen College Students Thinking and Debate Contest*
- 2017 Third Place, Best Debater | *The 3rd Shenzhen College Students Thinking and Debate Contest*
- 2018 Nomination Prize | *National Scholarship Award (Top 1%)*
- 2018 First Class | *Merit Student Scholarship for Exceptional Performance (Top 5%)*
- 2018 Dean's Award for Academic Excellence in College *(Top 1%)*
- 2018 Third Prize | *National College Student Social Practice and Science Contest on Energy Saving & Emission Reduction*
- 2019 Provincial second prize | *"TI Cup" National Undergraduate Electronics Design Contest*
- 2019 Excellence award | *Guangdong Entrepreneurship and Innovation Competition*
- 2019 Fisrt Prize | *Falling Walls Lab Shenzhen*
- 2019 First Class | *Merit Student Scholarship for Exceptional Performance (Top 5%)*
- 2019 Nomination Prize | *National Scholarship Award (Top 1%)*
- 2020 Gold Award | *"Challenge Cup" Guangdong College Student Entrepreneurship Plan Competition*
- 2020 Bronze Award | *"Challenge Cup" Nationall College Student Entrepreneurship Plan Competition*
- 2020 Excellent Graduate in Southern University of Science and Technology *(Top 5%)*

EXTRACURRICULAR ACTIVITIES

Secretary of SUSTech Student Union Secretariat *Sep.2016 - Jun.2017*

- Organized school cultural activities and coordinated the mission of various departments

Senior Counselor for Freshmen of Shude College, SUSTech *Jun.2017 - Jun.2019*

- Advised 60 freshman students via a series of mini-lectures and activities

Debate Team Leader of SUSTech *Feb.2018 - Feb.2019*

- Delivered training for members in competitions and maintained communications with other organizations

Volunteer of Future Forum & Shenzhen Summit *May.2018*

Volunteer of The Asia-Pacific Regional Seminar on MOOCs for Higher Education *Jun.2018*

Monitor of Class of Mechanical Engineering Major, Grade 2016, SUSTech *Sep.2018 - Present*

- Established a collaborative and caring community culture, solved problems and arranged social activities

Volunteer of China Hardware Innovation Camp (CHIC2019) *Jul.2019*

- Launched by École Polytechnique Fédérale de Lausanne, EPFL
- Participated in the project "Heptabox" to improve the effectiveness of medication intake